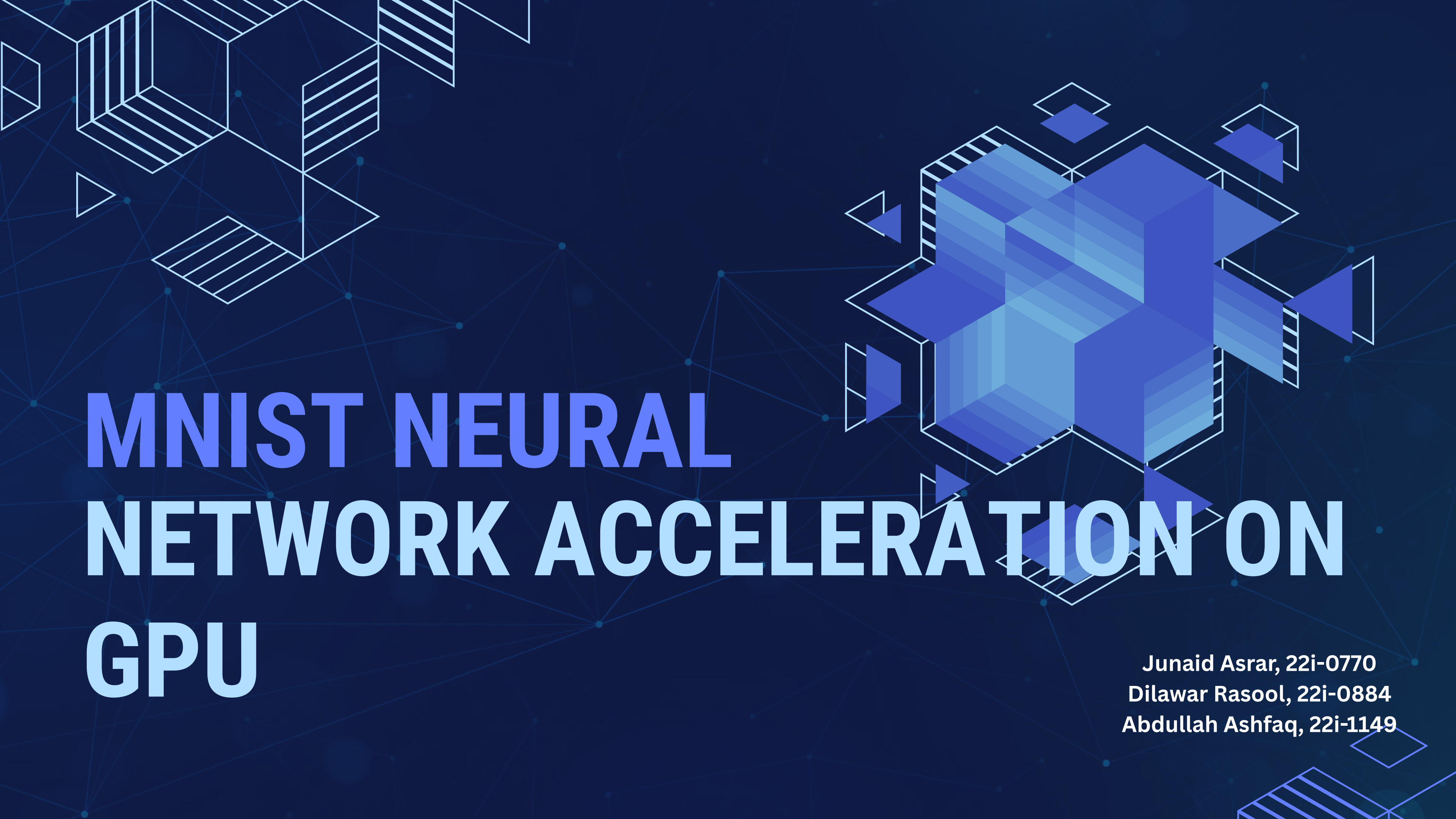




MNIST NEURAL NETWORK ACCELERATION ON GPU

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Project Overview

Objective: Accelerate MNIST digit recognition using GPU/CUDA.

- Dataset: 70,000 grayscale images (28x28 pixels).
- Versions Compared:
 - V1: Baseline CPU
 - V2: Naive GPU
 - V3: Optimized GPU
 - V4: Tensor Core GPU
 - Visual: Simple flowchart of versions (V1 → V4).

Performance Results

Versi on	Epoch 1 Time (s)	Epoch 2 Time (s)	Epoch 3 Time (s)	Total Training Time (s)	Test Accuracy (%)	Speedup Factor (x)
V1	7.587	7.559	7.665	22.811	96.77	1
V2	7.017	7.008	7.016	21.093	96.86	1.08
V3	4.925	4.925	4.925	14.828	96.90	1.54
V4	2.540	2.480	2.480	7.500	96.83	3

Key Performance Insights

V1

Baseline for comparison.

V2 (Naive GPU)

+8% speedup with
minimal changes.

V3 (Optimized GPU)

+54% speedup via
memory/launch
optimizations.

V4 (Tensor Core)

3x speedup leveraging
FP16/FP32 mixed
precision.

Approach for Each Version

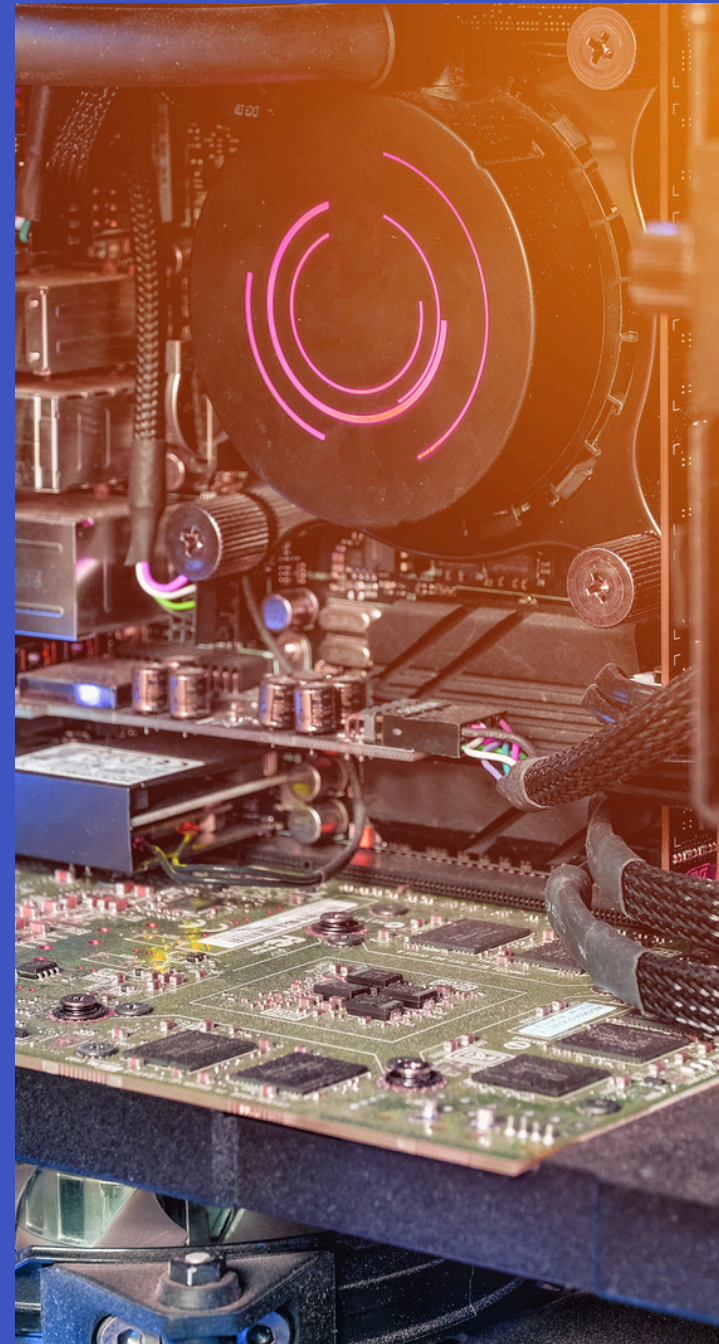
- V1 (CPU): NumPy, sequential execution.
- V2 (GPU): Basic CUDA kernels, no optimizations.
- V3 (Optimized GPU):
 - Tuned threads/blocks.
 - Shared memory, reduced CPU-GPU transfers.
- V4 (Tensor Core):
 - cuBLAS + Tensor Cores.
 - FP16/FP32 hybrid precision.



Optimizations in V3 & V4

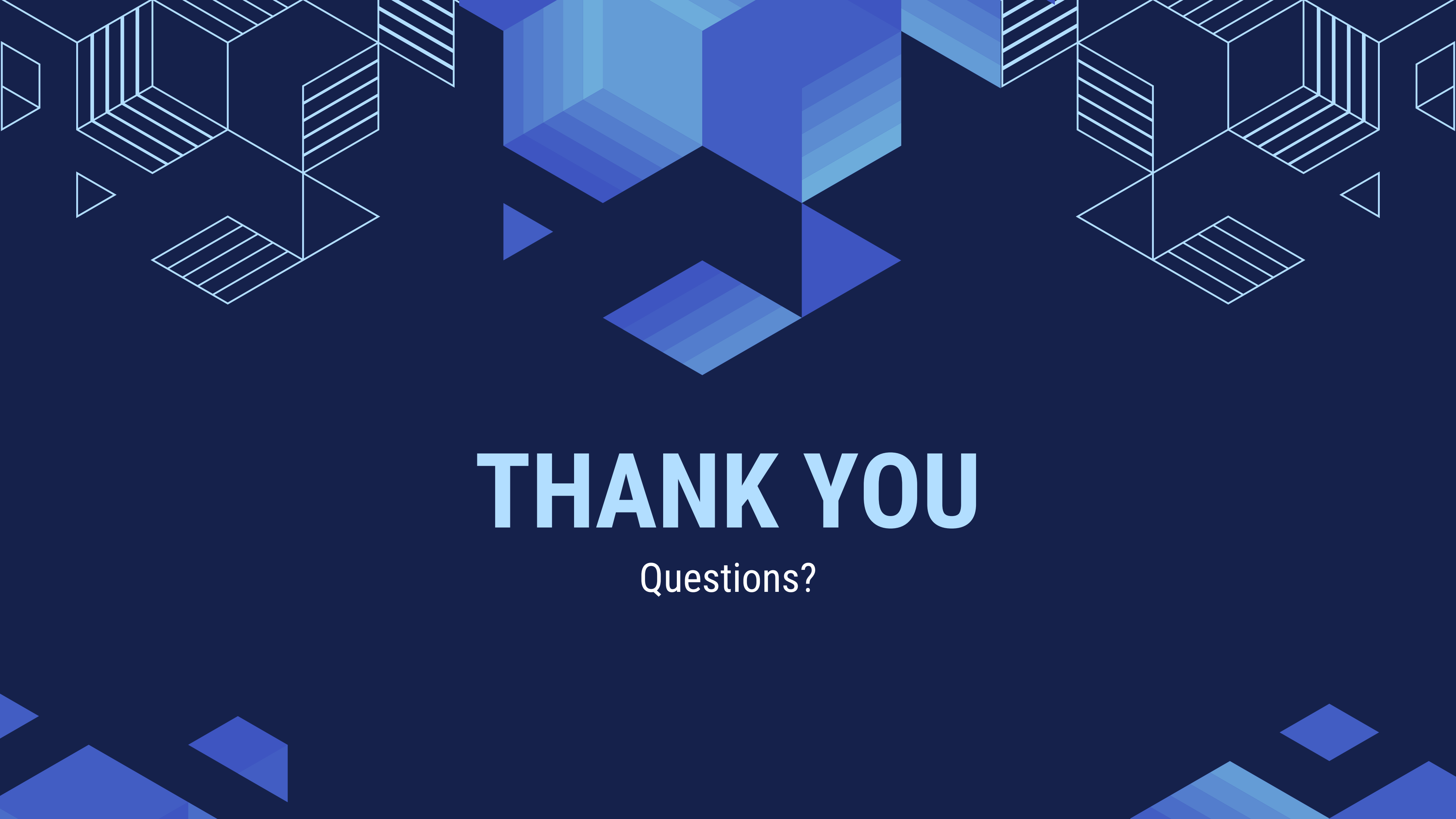
V3:

- Kernel launch configuration.
- Memory access optimization.
- V4:
- Tensor Core acceleration.
- Mixed-precision training.
- Visual: Side-by-side diagrams (GPU vs. Tensor Core).



Conclusion

- V4 (Tensor Core) achieved 67% faster training vs. V3.
- GPU optimizations (V3) + Tensor Cores (V4) = 3x total speedup.
- Accuracy remained stable (~96.8%).
- Visual: Summary infographic (key metrics + speedup).



THANK YOU

Questions?